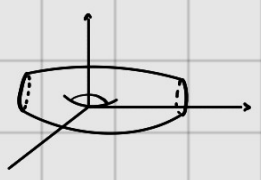
PRODOTTO SCALARE (NOTAZIONE)

$\langle \vec{v}, \vec{v}' \rangle = (\vec{v}, \vec{v}') = \vec{v} \cdot \vec{v}'$ ESEMPIO: $\langle (xy, 3-z, 2x), (2, 1-y^2) \rangle = 2xy + 1(3-z) + 2x(-y^2) = 2xy + 1 - z - 2xy^2$

$$\varphi: \begin{cases} x = (R + r\cos v)\cos u \\ y = (R + r\cos v)\sin u \\ z = r \sin v \end{cases} \qquad (u,v) \in [0, 2\pi] \times [0, 2\pi]$$



$(\Sigma_u) \quad \varphi_u = (-(R + r\sin v)\sin u, (R + r\cos v)\cos u, 0)$

$(\Sigma_v) \quad \varphi_v = (-r\sin v \cos u, -r \sin v \sin u, r\cos v)$

$$N = \varphi_u \wedge \varphi_v = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -(R + r\cos v)\sin u & (R + r\cos v)\cos u & 0 \\ -r\sin v \cos u & -r \sin v \sin u & r\cos v \end{vmatrix} = \left(r(R + r\cos v)\cos v \cos u, r(R + r\cos v)\sin v \cos u, r(R + r\cos v)(\sin^2 u \cdot \sin v + \cos^2 u \cdot \sin v) \right) =$$

$$\|N\|^2 = \|\varphi_u \wedge \varphi_v\|^2 = \| \left(r(R + r\cos v)\cos v \cos u, r(R + r\cos v)\sin v \cos u, r(R + r\cos v)\sin v \right) \|^2 = r^2 (R + r\cos v)^2 \left[\cos^2 v \cos^2 u + \sin^2 u \cdot \cos^2 v \right] =$$

$$= r^2 (R + r\cos v)^2 (\cos^2 v + \sin^2 v) = r^2 (R + r\cos v)^2 \implies \|N\| = r(R + r\cos v) > 0$$

$$\text{AREA}(\varphi) = \int_{\varphi} d\sigma = \iint_{\Omega} r(R + r\cos v)\, du\, dv = r \int_0^{2\pi} du \int_0^{2\pi} R + r\cos v \, dv = r \int_0^{2\pi} du \int_0^{2\pi} R + r\cos v \, dv = r \int_0^{2\pi} du \left[2\pi R + r(\sin 2\pi - \sin 0) \right]$$

$$= r 2\pi (2\pi R) = 4\pi^2 r R$$

$\Omega = \{ (u,v) \in [0, 2\pi] \times [0, 2\pi] \}$

$F(x,y,z) = \frac{1}{x^2 + y^2} \left(x, y, \frac{\sqrt{x^2 + y^2}}{r} \vec{z} \right)$ CALCOLARE IL FLUSSO DI F ATTRAVERSO φ

IL FLUSSO DI UN CAMPO F ATTRAVERSO UNA SUPERFICIE S È $\int_S \langle F, n_e \rangle d\sigma = \int_S \langle F, \frac{N}{\|N\|} \rangle d\sigma = \iint_{\Omega} \langle F, \frac{N}{\|N\|} \rangle \cdot \|N\| \, du\, dv = \iint_{\Omega} \langle F, N \rangle \, du\, dv$

NEL NOSTRO CASO: $\int_{\varphi} \langle F, n_e \rangle d\sigma = \iint_{\Omega} \left\langle \left(\frac{x}{x^2 + y^2}, \frac{y}{x^2 + y^2}, \frac{\sqrt{x^2 + y^2}}{r(x^2 + y^2)} \vec{z} \right), \right.$

SCRIVIAMO PRIMA F IN COORDINATE DELLA PARAMETRIZZAZIONE $F(u,v) = \left(\frac{(R + r\cos v)\cos u}{(R + r\cos v)^2 (\cos^2 u + \sin^2 u)}, \frac{(R + r\cos v)\sin u}{(R + r\cos v)^2 (\cos^2 u + \sin^2 u)}, \frac{\sin v}{R + r\cos v} \right)$

$F(x,y,z) = \frac{1}{x^2 + y^2} \left(x, y, \frac{\sqrt{x^2 + y^2}}{r} \vec{z} \right) \overset{\text{RISPETTO ALLA PARAMETRIZZAZIONE } \varphi}{=} \left(\frac{\cos u}{R + r\cos v}, \frac{\sin u}{R + r\cos v}, \frac{\sin v}{R + r\cos v} \right)$

ORA CALCOLIAMO IL FLUSSO

$$\int_{\varphi} \langle F, n_e \rangle d\sigma = \iint_{\Omega} \langle F, N \rangle \, du\, d\sigma = \iint_{\Omega} \langle \underbrace{\left(\frac{\cos u}{R + r\cos v}, \frac{\sin u}{R + r\cos v}, \frac{\sin v}{R + r\cos v} \right)}_{F}, \left(r(R + r\cos v)\cos v \cos u, r(R + r\cos v)\sin v \cos u, r(R + r\cos v)\sin v \right) \rangle \, du\, dv$$

$$= \iint_{\Omega} \left(r\cos^2 u \cos v + r\sin^2 u \cdot \cos v + r\sin^2 v \right) du\, dv = \iint_{\Omega} r\cos v + r\sin^2 v \, du\, dv = r \int_0^{2\pi} du \int_0^{2\pi} \cos v + \sin^2 v \, dv = 2\pi r \int_0^{2\pi} \cos v + \sin^2 v \, dv = 2\pi r \int_0^{2\pi} \sin^2 v \, dv =$$

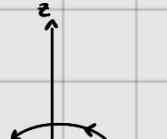
$$= 2\pi r \int_0^{2\pi} \frac{1 - \cos 2v}{2} dv = 2\pi r \cdot \frac{1}{2} \left[v \right]_0^{2\pi} = 2\pi^2 r$$

ESERCIZIO

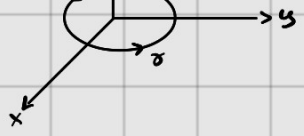
DATO IL CAMPO VETTORIALE $F(x,y,z) = (x^3, y^3, z^3)$, CALCOLARE L'INTEGRALE CURVILINEO $\oint_{\gamma} (F, T) ds$ DOVE γ È LA CIRCONFERENZA UNITARIA CONTENUTA NEL PIANO XY, CENTRATA NELL'ORIGINE ED ORIENTATA POSITIVAMENTE

CALCOLARE $\oint_{\gamma} \langle F, T \rangle ds = \int_{\Sigma} \langle \text{ROT} F, n_e \rangle d\sigma$ DOVE Σ È UNA SUPERFICIE CON $B\Sigma = \gamma$

AD ESEMPIO POSSIAMO SCEGLIERE

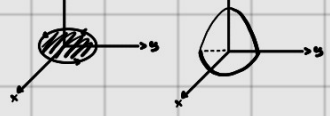






•) $\sum$ DISCO NEL PIANO xy

••) $\sum$ PARABOLOIDE $z = 1 - x^2 - y^2$



$$\text{ROT}F = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial x & \partial y & \partial z \\ x^3 & y^3 & z^3 \end{vmatrix} = (0,0,0) \qquad \oint_\gamma \langle F, T\rangle ds = \int_\Sigma \langle \text{ROT}F, n_e\rangle d\sigma = 0$$

•) $\sum : \begin{cases} x = \rho\cos\theta & \rho \in [0,1] \\ y = \rho\,\text{sen}\,\theta & \theta \in [0, 2\pi] \\ z = 0 \end{cases}$ $\left(\sum = \{x^2 + y^2 \le 1, z = 0\}\right)$

$n_e = (0,0,1)$ $\quad n_e = \sum_\rho \wedge \sum_\theta = (\underset{\text{DERIVATA}}{\cos\theta}, \text{sen}\,\theta, 0) \wedge (-\rho\,\text{sen}\,\theta, \rho\cos\theta, 0) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \cos\theta & \text{sen}\,\theta & 0 \\ -\rho\,\text{sen}\,\theta & \rho\cos\theta & 0 \end{vmatrix} = (0,0,\rho)$

DUE APPROCCI DIVERSI:

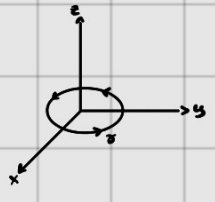
→ $\sum : \begin{cases} x = u \\ y = v \\ z = 0 \end{cases}$ $(u,v) \in \mathcal{D} = \{u^2 + v^2 \le 1\}$ $n_e = (0,0,1)$

Poi .. $\int_\Sigma \langle \text{ROT}F, n_e\rangle d\sigma = \iint_{\mathcal{D}} 0 \, du\,dv$ $\qquad n_e = \dfrac{N}{\|N\|} = \dfrac{\sum_u \wedge \sum_v}{\|\sum_u \wedge \sum_v\|}$

→ $\sum : \begin{cases} x = \rho\cos\theta \\ y = \rho\,\text{sen}\,\theta \\ z = 0 \end{cases}$ $(\rho,\theta) \in [0,1] \times [0, 2\pi] = \mathcal{D}'$ $n_e = (0,0,\rho)$

Poi ... $\int_\Sigma \langle \text{ROT}F, n_e\rangle d\sigma = \iint_{\mathcal{D}'} 0 \, d\rho\,d\theta$

$\int_\gamma \langle F, T\rangle ds$ $\quad \gamma : \begin{cases} x = \cos t \\ y = \text{sen}\,t \\ z = 0 \end{cases}$ $t \in [0, 2\pi]$



$T = (-\text{sen}\,t, \cos t, 0)$

$$\int_\gamma \langle F, T\rangle ds = \int_0^{2\pi} \langle (\cos^3 t, \text{sen}^3 t, 0^3), (-\text{sen}\,t, \cos t, 0)\rangle dt = \int_0^{2\pi} (-\cos^3 t + \text{sen}^3 t \cos t)\, dt = \left[\frac{\cos^4 t}{4} + \frac{\text{sen}^4 t}{4}\right]_0^{2\pi} = \boxed{0}$$

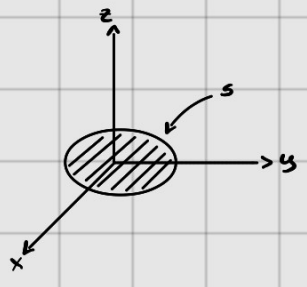
ESERCIZIO

ENUNCIARE IL TEOREMA DI STOKES. UTILIZZARLO PER CALCOLARE IL FLUSSO DEL ROTORE DI F ATTRAVERSO S, DOVE

$F(x,y,z) = (y + z^2)i + (x^2 + y)j + \text{ARCTg}\,(x^2 + y - z)k$ ED S È IL DISCO DEL PIANO $z = 0$ DI CENTRO L'ORIGINE E RAGGIO 1. CALCOLARE

SIA L'INTEGRALE SUPERFICIALE CHE QUELLO CURVILINEO

$F(x,y,z) = (y + z^2,\ x^2 + y,\ \text{ARCTg}\,(x^2 + y - z))$ $\qquad BS : \begin{cases} x = \cos t \\ y = \text{sen}\,t \\ z = 0 \end{cases}$ $t \in [0, 2\pi]$



$T = (-\text{sen}\,t, \cos t, 0)$ $\qquad \int_{BS} \langle F, T\rangle ds = \int_S \langle \text{ROT}F, n_e\rangle d\sigma$

$$\int_{BS} \langle F, T\rangle ds = \int_0^{2\pi} \langle (\text{sen}\,t + 0^2,\ \cos^2 t + \text{sen}\,t,\ \text{ARCTg}\,(\cos^2 t + \text{sen}\,t - 0)), (-\text{sen}\,t, \cos t, 0)\rangle dt =$$

$$= \int_0^{2\pi} -\text{sen}^2 t + (\cos^2 t + \text{sen}\,t)\cos t + 0\, dt = \int_0^{2\pi} -\text{sen}^2 t + \cos^3 t + \text{sen}\,t\cos t\, dt = \int_0^{2\pi} \cos^2 t - 1 + \cos t\,(1 - \text{sen}^2 t)\, dt =$$

$$= \int_0^{2\pi} \frac{1 - \cos 2t}{2} - 1 + \cos t - \cos t\,\text{sen}^2 t\, dt = \int_0^{2\pi} -\frac{1}{2} - \frac{\cos t}{2} - \cos t\,\text{sen}^2 t\, dt = \left[-\frac{1}{2}t - \frac{\text{sen}\,2t}{4} - \frac{\text{sen}^3 t}{3}\right]_0^{2\pi} = \boxed{-\pi}$$

ALTRO MODO PER SVOLGERE L'ESERCIZIO:

$$\Sigma:\begin{cases}x=u\\ y=v\\ z=0\end{cases} \quad (u,v)\in\Omega=\{x^2+y^2\le 1\} \quad \Sigma_u=(1,0,0) \quad \Sigma_v=(0,1,0) \quad N=\Sigma_u\wedge\Sigma_v=(1,0,0)\wedge(0,1,0)=(0,0,0)$$

$$N=\Sigma_u\wedge\Sigma_v=\begin{vmatrix} i & j & k\\ 1 & 0 & 0\\ 0 & 1 & 0\end{vmatrix}=(0\cdot 0-1\cdot 0,\ 1\cdot 0-0\cdot 0,\ 1\cdot 1-0\cdot 0)=[0,0,1] \quad N=(0,0,1)$$

$F_x=y+z^2 \quad F_y=x^2+y \quad F_z=\text{ARCtg}(\underbrace{x^2+y-z}_{v})$

$$\text{ROT } F=\begin{vmatrix} i & j & k\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ \frac{\partial F}{\partial x} & \frac{\partial F}{\partial y} & \frac{\partial F}{\partial z}\end{vmatrix}=\left[\frac{\partial F_z}{\partial y}-\frac{\partial F_y}{\partial z},\ \frac{\partial F_z}{\partial x}-\frac{\partial F_x}{\partial z},\ \frac{\partial F_y}{\partial x}-\frac{\partial F_x}{\partial y}\right]$$

$$=\begin{vmatrix} i & j & k\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ y+z^2 & x^2+y & \text{ARCtg}(x^2+y-z)\end{vmatrix}=\left[\frac{1}{1(x^2+y-z)^2}-0,\ \frac{2x}{1(x^2+y-z)^2}-2z,\ 2x-1\right]$$

$$\frac{\partial F_z}{\partial y}=\text{ARCtg}(v)\cdot 1=\frac{1}{1+v^2}\cdot 1=\frac{1}{1(x^2+y-z)^2}\cdot 1$$

$$\frac{\partial F_y}{\partial z}=0 \quad \frac{\partial F_x}{\partial z}=2z \quad \frac{\partial F_y}{\partial x}=2x \quad \frac{\partial F_x}{\partial y}=1$$

$$\frac{\partial F_z}{\partial x}=\text{ARCtg}(v)\cdot 2x=\frac{1}{1+v^2}\cdot 2x=\frac{2x}{1(x^2+y-z)^2}$$

$n_e=(0,0,1)$

COORDINATE POLARI $H\begin{cases}u=\rho\cos\theta & \rho\in[0,1]\\ v=\rho\sin\theta & \theta\in[0,2\pi]\end{cases}$ $\quad dudv=d\rho d\theta$

$$\Sigma:\begin{cases}x=u\\ y=v\\ z=0\end{cases}$$

sostituisci i valori

$$\int_\Sigma \langle \text{ROT}F, n_e\rangle d\sigma=\iint_\Omega\left\langle\left(\frac{1}{1(u^2+v^2)^2},\frac{2u}{1(u^2+v^2)^2},2u-1\right),(0,0,1)\right\rangle dudv=\iint_\Omega(0+0+2u-1)dudv=\iint_\Omega(2u-1)dudv=$$

$$=\iint_{\Omega_H}(2\rho\cos\theta-1)\underbrace{\rho}_{\det(H)}d\rho d\theta=\iint_{\Omega_H}(2\rho^2\cos\theta-\rho)d\rho d\theta=\int_0^1 d\rho\int_0^{2\pi}2\rho^2\cos\theta-\rho\,d\theta=\int_0^1-2\pi\rho\,d\rho=-\pi\int_0^1 2\rho\,d\rho=-\pi\left[\rho^2\right]_0^1=-\pi$$

ESERCIZIO

CALCOLARE $\int_{+\partial\Sigma}\langle F,T\rangle ds$ DOVE $F=(x^2,xy,y+2z)$ E Σ È LA PORZIONE DI PARABOLOIDE $z=x^2+y^2$, $1\le z\le 9$. CALCOLARE SIA L'INTEGRALE SUPERFICIALE CHE QUELLO CURVILINEO

$$\Sigma=\{z=x^2+y^2,\ 1\le z\le 9\} \quad F=(x^2,xy,y+2z) \quad \text{ROT } F=\begin{vmatrix} i & j & k\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ z^2 & xy & y+2z\end{vmatrix}=(1-0,\ 2z-0,\ y-0)=(1,2z,y)$$

$$\Sigma:\begin{cases}x=u\\ y=v\\ z=u^2+v^2\end{cases} \quad (u,v)\in\Omega=\{1\le x^2+y^2\le 9\} \quad n_e=\left(\frac{\partial F}{\partial x},\frac{\partial F}{\partial y},\frac{\partial F}{\partial z}\right)=(-2u,-2v,1) \quad ds=\sqrt{4u^2+4v^2+1}\,dxdy$$

$z=x^2+y^2$

$$\int_\Sigma\langle F,n_e\rangle d\sigma=\iint_\Omega\langle(1,2(u^2+v^2),v),(2u,-2v,1)\rangle dudv=\iint_\Omega -2u-4v(u^2+v^2)+v\,dudv$$

COORDINATE POLARI $\begin{cases}u=\rho\cos\theta & \rho\in[1,3]\\ v=\rho\sin\theta & \theta\in[0,2\pi]\end{cases}$

$$=\int_0^{2\pi}d\theta\int_1^3-2\rho\cos\theta-4\rho\sin\theta(\underbrace{\rho^2\cos^2\theta+\rho^2\sin^2\theta}_{1})+\rho\sin\theta\,\rho\,d\rho=\int_0^{2\pi}d\theta\int_1^3-2\rho\cos\theta-4\rho^3\sin\theta+\rho\sin\theta\,\rho\,d\rho=$$

$$=\int_0^{2\pi}d\theta\int_1^3-2\rho\cos\theta-4\rho^3\sin\theta+\rho^2\sin\theta\,d\rho=\int_0^{2\pi}\cos\theta+\sin\theta+\sin\theta\,d\theta\int_1^3-2\rho-4\rho^3+\rho^2\,d\rho=$$

$\int_0^{2\pi}\cos\theta\,d\theta=0 \quad \int_0^{2\pi}\sin\theta\,d\theta=0$

$$=\int_0^{2\pi}\cos\theta+\sin\theta+\sin\theta\,d\theta\left[-\frac{2\rho^2}{2}-\frac{4\rho^4}{4}+\frac{\rho^3}{3}\right]_1^3=0$$

SIA L'INTEGRALE SUPERFICIALE, SIA L'INTEGRALE CURVILINEO DÀ COME RISULTATO ZERO

ESERCIZIO